

**Reputability and Quantity of Misinformation.** The next proposition shows that greater reliability of articles and less misinformation lead to uniformly more sharing. We define greater reliability as a first-order stochastically dominant shift of the reliability distribution of articles from  $F$  to  $F'$  (where  $F' \succeq_{FOSD} F$ ). We refer to less misinformation as a shift of the function  $\phi$  to some  $\phi' \geq \phi$ .

**Proposition 1.** *More reliability of articles and/or less misinformation lead to uniformly more sharing.*

These results are intuitive. Holding constant the degree of misinformation, a shift towards more reliable articles reduces the relevant vector of thresholds  $(b^*, b^{**})$ , and this means that each agent becomes more likely to share. Holding constant the distributional reliability, less misinformation reduces the cost associated with sharing, triggering more aggressive sharing by all agents.

The predictions summarized in Proposition 1 are consistent with the existing evidence. Pennycook and Rand (2019a) document that social media users recognize low-reliability content sources (e.g., Breitbart or Infowars), and this content is not typically shared by attentive social media users, regardless of partisanship (see Pennycook and Rand (2019b)). This proposition also clarifies that viral spread of misinformation is not a mechanical effect in our model: if anything, less reliable articles that are more likely to contain misinformation are less likely to become viral. We will see that other aspects of social media interactions—sensational content and the nature of the sharing network—are often responsible for viral spread of misinformation.

**Sensational Content and Reputational Concerns.** Let us next consider two key parameters of users' utility. The first is  $\kappa$ , which we assume captures how *sensational* the article is. Higher  $\kappa$  implies that agents receive greater value from future shares, because these shares are associated with others paying more attention or perhaps being entertained more by the relevant posts. This greater utility is independent of the content's veracity. As an example, headlines such as “Trump Pays \$2 Million to 8 Charities for Misuse of Foundation” are more sensational than headlines like: “108,000 people will get medical debt relief after Stacey Abrams' PAC gifts \$1.34 million.”<sup>12</sup> We think of  $\kappa$  varying at the level of articles, so that some articles will be more sensational than others.

The second parameter is  $d$ , standing in for the importance of *reputational concerns*. Higher  $d$  means that dislikes are more damaging, which corresponds to the agent being more concerned about receiving many dislikes. We think of  $d$  as varying at the level of communities (certain communities of users, for example, academics, may have more reputational concerns).

**Proposition 2.** *More sensational articles and/or lower reputational concerns of users lead to uniformly more sharing.*

The results in this proposition are both intuitive and immediate. Nevertheless, they have important applications. First, the results of this proposition cohere with the patterns documented in Duffy et al. (2020), suggesting that social media participants often share a story that is “too good not to share”, and do so even when they realize it is also “too good to be true”. Second, the link between reputational concerns and misinformation is consistent with the evidence that in settings where reputation matters

<sup>12</sup>Sources are, respectively, <https://www.nytimes.com/2019/12/10/nyregion/trump-foundation-lawsuit-attorney-general.html> and <https://www.npr.org/2021/10/27/1049675704/stacey-abrams-medical-debt-relief-pac-donation>.